CHAPTER 6

BODY FLUIDS AND CIRCULATION

CHAPTER ESSENCE AT A GLANCE:
Rhythmic cardiac cycle dynamics, double circulation circuits, and electrocardiographic electrical conduction.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Blood Elements, ABO Groups, and Rh Incompatibility:** Formed elements comprise erythrocytes (biconcave, transport O2), leucocytes (granulocytes and agranulocytes for immune defense), and platelets (thrombocytes for clotting). ABO grouping is governed by surface antigens A and B. Rh incompatibility (Erythroblastosis foetalis) occurs when an Rh-negative mother carries an Rh-positive fetus, prevented by administering anti-Rh antibodies immediately following the first delivery.

• **Cardiac Conduction and the Cardiac Cycle:** Nodal rhythmicity: SA node (pacemaker generating 70 to 75 action potentials per minute) initiates impulses across atria to AV node, Bundle of His, and Purkinje fibers. The cardiac cycle (0.8 s) comprises atrial systole (0.1 s), ventricular systole (0.3 s), and joint diastole (0.4 s). Each ventricle pumps approximately 70 mL stroke volume per beat, yielding Cardiac Output = $SV imes HR ■pprox 5 ext{ liters per minute}$.

• **Electrocardiogram (ECG) Deflection Waves:** A standard ECG records cardiac electrical activity: P wave represents atrial depolarization (initiating atrial systole); QRS complex represents ventricular depolarization (initiating ventricular contraction); T wave represents ventricular repolarization (return from excited to resting state). Counting QRS complexes in a given time interval determines heart rate.

• **Double Circulation and Cardiovascular Pathology:** Humans possess complete double circulation: Pulmonary circulation (Right ventricle to pulmonary artery to lungs to pulmonary veins to left atrium) and Systemic circulation (Left ventricle to aorta to systemic tissues to vena cava to right atrium). Hepatic portal system carries blood from intestine to liver before systemic return. Disorders: Hypertension (>140 over 90 mm Hg), Atherosclerosis (CAD), Angina pectoris, and Heart failure.

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{Cardiac Output} = \text{Stroke Volume (70 mL)} \times \text{Heart Rate (72 bpm)} \approx 5.04 \text{ L min}^{-1}$$

Cardiac Output Determination via Stroke Volume and Heart Rate

$$\text{ECG Waves} = \text{P (Atrial Depol)} + \text{QRS (Ventricular Depol)} + \text{T (Ventricular Repol)}$$

Electrocardiographic Vector Waves of the Cardiac Cycle

$$\text{Circulatory Loops} = \frac{\text{Pulmonary: RV} \rightarrow \text{Pulmonary Artery} \rightarrow \text{Lungs} \rightarrow \text{LA}}{\text{Systemic: LV} \rightarrow \text{Systemic Aorta} \rightarrow \text{Tissues} \rightarrow \text{RA}}$$

Complete Closed Double Circulation Pathways

$$\text{Coagulation Cascade} = \text{Prothrombin} \xrightarrow[Ca^{2+}]{\text{Thrombokinase}} \text{Thrombin} \longrightarrow \text{Fibrinogen} \rightarrow \text{Fibrin Clot}$$

Enzymatic Cascade of Blood Clotting

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Heart failure (inability of the heart to pump blood effectively to meet metabolic needs) is distinct from cardiac arrest (heart stops beating) and heart attack (myocardial infarction).

• **Alert:** Erythroblastosis foetalis occurs only when an Rh-negative mother is carrying a second Rh-positive child; it is completely prevented by administering RhoGAM anti-Rh antibodies.

• **Alert:** The SA node is the natural pacemaker because it generates the highest frequency of action potentials (70 to 75 per minute) and initiates rhythmic contraction.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is Erythroblastosis foetalis? State the medical preventative measure administered to an Rh-negative mother. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Erythroblastosis foetalis: A severe hemolytic disease of the newborn occurring when an Rh-negative mother carries an Rh-positive fetus; maternal anti-Rh antibodies enter fetal circulation, destroying fetal RBCs and causing severe anemia and jaundice. **[1 Mark]**

• Preventative measure: Administering anti-Rh antibodies (RhoGAM) to the Rh-negative mother immediately following the delivery of the first Rh-positive child. **[1 Mark]**

2. Distinguish between atherosclerosis (coronary artery disease) and angina pectoris. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Atherosclerosis (CAD): Caused by deposition of calcium, fat, cholesterol, and fibrous tissue in arteries that supply the heart musculature, narrowing the lumen. **[1 Mark]**

• Angina pectoris: Acute chest pain that appears when insufficient oxygen reaches the heart muscle due to reduced coronary blood flow. **[1 Mark]**

3. Draw a standard electrocardiogram (ECG) waveform, label the P wave, QRS complex, and T wave, and state the physiological event represented by each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• P wave: Represents the electrical excitation or depolarization of the atria, which leads to the contraction of both atria. **[1 Mark]**

• QRS complex: Represents the depolarization of the ventricles, which initiates ventricular contraction beginning shortly after Q. **[1 Mark]**

• T wave: Represents the repolarization of the ventricles, signifying the return of ventricles from excited to normal resting state (end of T wave marks end of systole). **[1 Mark]**

4. Explain the sequential mechanical events during a single cardiac cycle, specifying the duration and state of heart chambers during atrial systole, ventricular systole, and joint diastole. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Joint diastole (0.4 s): All four chambers of the heart are in a relaxed state; tricuspid and bicuspid valves are open, allowing blood from pulmonary veins and vena cava to flow into left and right ventricles. **[1 Mark]**

• Atrial systole (0.1 s): SA node fires an action potential, causing both atria to contract simultaneously, increasing blood flow into ventricles by about 30 percent. **[1 Mark]**

• Ventricular systole (0.3 s): AV node conveys impulse through bundle of His and Purkinje fibers; ventricles contract, intraventricular pressure closes tricuspid and bicuspid valves (first heart sound lub) and opens semilunar valves, pumping blood into aorta and pulmonary artery (second heart sound dub occurs on ventricular relaxation). **[1 Mark]**

5. If an endurance athlete has a resting heart rate of 50 beats per minute and a stroke volume of 100 mL, calculate the athlete's cardiac output and compare it with that of an average sedentary adult. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Calculation: Cardiac Output = Stroke Volume * Heart Rate = 100 mL * 50 beats per minute = 5000 mliters per minute = 5.0 liters per minute. **[1.5 Marks]**

• Comparison: An average adult has a heart rate of 72 bpm and a stroke volume of 70 mL (70 * 72 = 5040 mliters per minute); the athlete achieves the same 5.0 liters per minute cardiac output with greater cardiovascular efficiency (higher stroke volume, lower heart rate). **[1.5 Marks]**

6. Trace the pathway of blood flow through the human double circulation, distinguishing between the pulmonary circuit and the systemic circuit. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Pulmonary circuit: Deoxygenated blood pumped from right ventricle enters the pulmonary artery, passes through lungs for oxygenation, and returns via pulmonary veins into the left atrium. **[1.5 Marks]**

• Systemic circuit: Oxygenated blood pumped from left ventricle enters the aorta, is distributed through arteries and capillaries to body tissues, and deoxygenated blood is collected by vena cava into the right atrium. **[1.5 Marks]**

7. A patient's ECG trace reveals completely absent P waves and an irregular rhythm of QRS complexes. Identify which chamber of the heart has compromised rhythmic excitation, specify the defective nodal tissue, and explain why cardiac output is reduced. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Defective chamber: The atria are failing to undergo coordinated electrical depolarization and contraction (atrial fibrillation). **[1 Mark]**

• Defective nodal tissue: The Sinoatrial (SA) node is malfunctioning, failing to generate regular rhythmic pacemaker action potentials. **[1.5 Marks]**

• Reduction in cardiac output: Atrial contraction normally tops up ventricular filling by about 30 percent; in the absence of organized atrial systole, end-diastolic ventricular volume drops, leading to decreased stroke volume and reduced cardiac output. **[1.5 Marks]**